Results — Flow-Pattern Anomaly Detection at the IoT Gateway

Madhav Dogra · 2026-06-15 · IoT-23 `CTU-IoT-Malware-Capture-34-1` (Mirai)

Setup

Data. IoT-23 scenario 34-1 pcap (120.5 MB, 233,865 packets) → **4,662 flows**, each the first K=32 packets × 13 payload-free features; labels joined from the Zeek `conn.log.labeled` by 5-tuple. Classes: Mirai 4055, Benign 290, DDoS 211, Recon 106. Split 70/15/15 (test 700).
Model. Mamba encoder (d=128, 4 layers) + softmax classifier + one-class deep-SVDD anomaly head.
Run. 3 stages × 8 epochs, CPU (Apple M3 Pro). Extraction ≈ 1.5 s; **training 659 s (≈11 min)**; inference **5.1 ms/flow**.

Results

Metric	Value
Accuracy	**96.3 %**
Macro-F1 (4 classes)	**0.793**
Weighted-F1	0.963
Anomaly ROC-AUC	**0.994**
Anomaly PR-AUC	0.999
Latency p50/p95/p99	5.1/5.5/6.0 ms

Confusion matrix (rows true, cols pred)

	B	D	R	M
Benign (B)	47	0	3	0
DDoS (D)	2	19	**21**	0
Recon (R)	0	0	18	0
Mirai (M)	0	0	0	590

Class	P	R	F1	N
Benign	0.959	0.940	0.949	50
DDoS	1.000	0.452	0.623	42
Recon	0.429	1.000	0.600	18
Mirai	1.000	1.000	1.000	590
Macro avg	0.847	0.848	**0.793**	700
Weighted avg	0.982	0.963	0.963	700

Takeaway.

- Benign-trained anomaly head detects attacks at **ROC-AUC 0.994** on real malware traffic — the zero-day capability.
- 96.3 % accuracy; honest macro-F1 0.793 (single, Mirai-heavy capture).
- Main error: DDoS read as Recon (similar short-packet patterns); both classes are small.

First real-data run, single scenario. Adding more IoT-23 captures (class balance) and per-class focal weights are the next steps. Synthetic-data runs omitted (trivially separable by construction).